

MORPH — Core-MLP Lifecycle: MORTAR · CMS Pruning · ReMoE

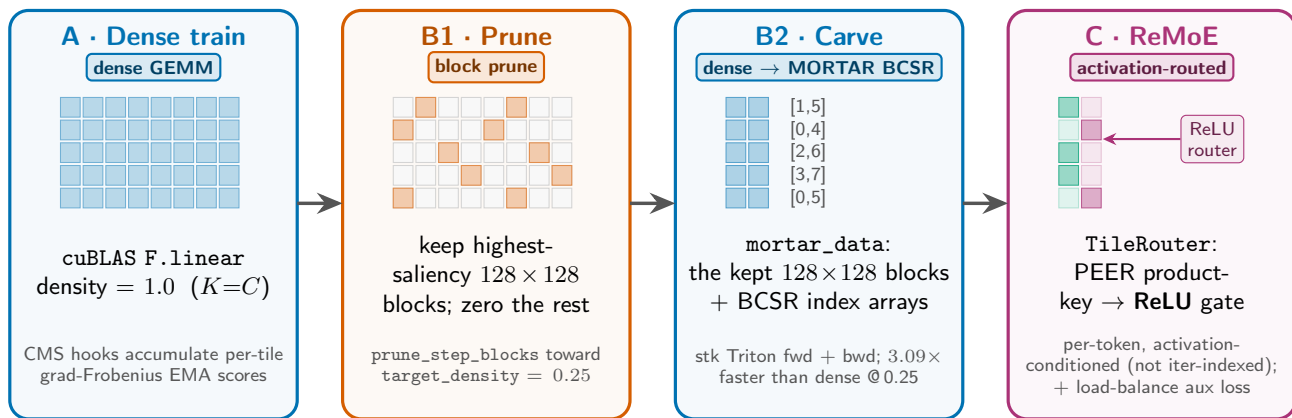
training step →

step 0

prune_start

compact_step

route_start



MORTAR BCSR: a $[d_{\text{out}}, d_{\text{in}}]$ weight tiles into $R_b \times C_b$ blocks of 128×128 ($R_b = d_{\text{out}}/128$, $C_b = d_{\text{in}}/128$); pruning keeps nnz of them, so **density** = $\text{nnz}/(R_b C_b)$. **Scope:** all SwiGLU MLPs (gate_up + down) across **prelude + core + coda** (MORTAR-only — no scope knob); the core block is re-applied every loop iteration.

CMS schedule: accumulate saliency EMA every step · **prune_step_blocks** from **prune_start** toward **target_density** · **carve()** once at **compact_step** · ReMoE at **route_start**. Saliency = **score_mode** (taylor / grad / magnitude) EMA, pooled 16×16 tile → 128×128 block (score at tile, execute at block).

ReMoE: routing is *per-token activation-conditioned* on each MORTAR layer (prelude / core / coda). In the looped core, different iterations route differently only because their activations differ — *not* via any iteration-index input.